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| <b>Session 1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Start of the Lesson (10 minutes)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p>Answer the following prompts individually, then discuss your responses with your group. Be ready to share one key insight during class discussion.</p> <p>1. What part of your prototype are you most confident about testing, and why?<br/><u>We are most confident in testing the personalized supplement recommendation feature of our prototype because it is the main function of the system.</u></p> <p>2. What research or evidence supports your design and technology choices?<br/><u>Research studies by Alowais et al. (2023), Bush et al. (2021), and Sonboli et al. (2021) support the use of AI and recommendation systems in healthcare and nutrition.</u></p> <p>3. Why is it important to test your prototype internally first (UAT or Alpha Testing) before presenting it to external users or evaluators?<br/><u>Internal testing is important because it helps identify bugs, usability problems, and missing features before the system is presented to actual users or evaluators.</u></p> <p>4. How can your team use feedback to improve your system while keeping it aligned with user needs and ethical standards?<br/><u>Our team can use feedback by analyzing the suggestions and concerns of testers to improve the accuracy, usability, accessibility, and reliability of the system.</u></p> |

### Try It Out: Why Test Internally First?

#### 1. What could happen if a prototype is released or presented without internal testing?

If a prototype is released without internal testing, users may encounter bugs, inaccurate recommendations, poor navigation, or system crashes. This can reduce user trust and negatively affect the credibility of the project.

#### 2. Why is it important to identify bugs and usability issues early?

It is important to identify bugs and usability issues early because fixing problems during the early stages is easier, faster, and less costly. Early testing also improves the quality and reliability of the system.

#### 3. How can gathering structured feedback help improve your project quality and teamwork?

Structured feedback helps the team identify strengths and weaknesses of the system clearly. It also improves teamwork by helping members communicate problems, assign responsibilities, and prioritize improvements effectively.

### Let's Practice: Conducting Internal Testing and Feedback Analysis (60 minutes)

**Objective:** Apply testing practices to validate your prototype (built in Module 6 and justified in Module 7) by conducting UAT and Alpha Testing, collecting structured feedback, and creating a revision plan.

#### Step 1: Create Your UAT Script

##### Instructions:

1. Work with your group to write **test cases** for your prototype.
2. Each test case should include:
  - **Task to test**
  - **Test data** (if applicable)
  - **Expected result** (acceptance criteria)

##### Blank UAT Test Case Template

| TC ID | Task                        | Test Data                                         | Expected Result                                                |
|-------|-----------------------------|---------------------------------------------------|----------------------------------------------------------------|
| TC-01 | User Registration / Sign Up | Email: test@email.com,<br>Password: Password123   | Account created successfully;<br>user redirected to dashboard. |
| TC-02 | Login with wrong password   | Email: test@email.com,<br>Password: wrongpassword | Access denied; error message<br>"Invalid credentials" appears. |

#### Step 2: Conduct Alpha Testing

##### Instructions:

1. **Swap prototypes** with another group for internal testing.
2. Invite **2–3 testers** from other groups to test your prototype using your UAT Script.
3. Testers and observers record:
  - **Pass/Fail results**
  - **Errors or bugs found**
  - **User reactions** (ease, confusion, satisfaction)

##### Sample of Feedback Collection Template:

| Task Tested         | Success (Y/N) | Issues Encountered      | Tester Feedback         | Suggested Fix       |
|---------------------|---------------|-------------------------|-------------------------|---------------------|
| Add new task        | Y             | None                    | "Worked smoothly"       | Keep as is          |
| Add task (no title) | Y             | Error message appeared  | "Good validation"       | Maintain validation |
| View overdue tasks  | N             | Overdue not highlighted | "Hard to track urgency" | Add badge/flag      |

##### Blank Feedback Collection Template

| Task Tested                 | Success (Y/N) | Issues Encountered                      | Tester Feedback                                                  | Suggested Fix                                   |
|-----------------------------|---------------|-----------------------------------------|------------------------------------------------------------------|-------------------------------------------------|
| User Registration / Sign Up | Y             | None                                    | "Registration form loads quickly and works fine."                | Keep as is.                                     |
| Login with wrong password   | Y             | Error text color is gray instead of red | "The validation works, but the error message is hard to notice." | Change text color to bright red for visibility. |

### Step 3: Analyze Feedback

#### Instructions:

1. Review the results of your Alpha Testing.
2. Summarize findings in the Feedback Analysis Table.
3. Identify what worked well and what needs improvement.
4. Rank improvements by priority (High, Medium, Low).

#### Sample Feedback Analysis Template

| Feedback/Observation       | Positive Insights | Issues/ Challenges | Suggested Improvements | Priority (H/M/L) |
|----------------------------|-------------------|--------------------|------------------------|------------------|
| Add task works fine        | Fast, easy        | None               | Keep as is             | L                |
| Validation prevents errors | Helpful check     | None               | Keep as is             | L                |
| Overdue tasks not flagged  | Sorting works     | Urgency missing    | Add red badge/flag     | H                |

#### Blank Feedback Analysis Template

| Feedback/Observation       | Positive Insights           | Issues/ Challenges | Suggested Improvements | Priority (H/M/L) |
|----------------------------|-----------------------------|--------------------|------------------------|------------------|
| Registration is functional | Fast and easy system access | None               | Keep as is.            | L                |

|                              |                                     |                                           |                                                |   |
|------------------------------|-------------------------------------|-------------------------------------------|------------------------------------------------|---|
| Login validation looks plain | Effectively blocks wrong passwords. | Error message doesn't stand out visually. | Style error text to red and add an alert icon. | M |
|------------------------------|-------------------------------------|-------------------------------------------|------------------------------------------------|---|

## Session 2

### Start of the Lesson (10 minutes)

Answer the following prompts individually, then discuss your responses with your group. Be ready to share one key insight during class discussion.

1. What was the main purpose of conducting UAT and Alpha Testing in the previous session?

The main purpose of conducting UAT and Alpha Testing was to evaluate whether the system functions correctly, meets user requirements, and provides a user-friendly experience before being presented to external users.

2. What was the most important feedback your team received during testing?

The most important feedback received was that the personalized recommendation feature needed more accurate filtering based on users' health conditions and dietary needs.

3. How did your team decide which issues or improvements to prioritize after analyzing feedback?

Our team prioritized issues based on their impact on user experience, system functionality, and reliability.

4. Why is it important to document testing results and revision plans before continuing to the next prototype build?

Documenting testing results and revision plans helps the team track issues, organize improvements, and ensure that future updates are based on evidence and user feedback.

## Try It Out: Why Do We Revise?

### 1. What kind of feedback did your Alpha Testing reveal about your prototype's usability or performance?

Alpha Testing revealed feedback related to mobile responsiveness, recommendation accuracy, system speed, and user interface readability.

### 2. Which of those findings were most critical to address first, and why?

The most critical findings were the recommendation accuracy and mobile responsiveness issues because they directly affect the user experience and the reliability of the system.

### 3. How can clear documentation of revisions improve communication within your team and with your mentor or evaluators?

Clear documentation helps the team communicate responsibilities, progress, and improvements effectively. It also allows mentors and evaluators to understand how feedback was analyzed and applied to improve the project.

#### Let's Practice: Draft a Revision Plan and Alpha Test Report (40 minutes)

**Objective:** Develop a structured plan for your top prototype improvements, supported by evidence from your Alpha Testing results.

##### Part A: Draft Your Revision Plan

###### Instructions:

1. Review your Feedback Analysis Table from Session 1.
2. Select the top 2–3 improvements that will make the biggest difference to your prototype.
3. Record them in the table below.
4. Discuss and assign roles, timelines, and priorities.

##### Sample Revision Plan

| Issue/Challenge             | Suggested Fix              | Owner        | Timeline        | Priority |
|-----------------------------|----------------------------|--------------|-----------------|----------|
| Overdue tasks not flagged   | Add CSS highlight          | UI Dev       | By next session | H        |
| Cancel button saves changes | Fix cancel logic           | Frontend Dev | By next session | H        |
| Real-time sync slow (4s)    | Optimize Firebase listener | Backend Dev  | 2 weeks         | M        |

##### Blank Revision Plan Template

| Issue/Challenge                                                          | Suggested Fix                                                                      | Owner       | Timeline        | Priority |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------|-----------------|----------|
| AI analysis outputs are sometimes inaccurate or inconsistent.            | Refine backend prompt engineering and add strict validation logic to data parsing. | Backend Dev | 1 week          | H        |
| Validation error messages during bad logins look plain and unnoticeable. | Style error text to a bright red color and introduce a clear warning alert icon.   | UI Dev      | By next session | M        |

## Part B: Short Alpha Test Report

**Key Findings:** The user interface and registration flow are stable and responsive, but internal testing revealed that the AI analysis features occasionally output inaccurate or inconsistent results.

**Critical Issues:** The AI analysis inaccuracy must be fixed first, as it directly impacts the core functionality and reliability of the system.

**Planned Improvements:** The backend developer will refine prompt engineering and implement strict data validation, while the frontend team will optimize dashboard loading and improve visual error alerts.

### Questions to Guide Your Report:

**1. What evidence (feedback) supports your chosen improvements?** Alpha testing logs explicitly showed instances where the system's generated data did not match expected outcomes.

**2. Which issue is the most critical to fix before Beta Testing?** Improving the accuracy of the AI analysis is the most critical fix so we don't give external testers flawed data.

**3. How does your testing feedback confirm or challenge your Module 7 Tech Rationale?** It confirms our React/Node.js stack is fast and stable, but challenges our backend processing logic, showing we need tighter data validation rules.

#### Think Back and Think About (10 minutes)

**Think Back:** How did conducting **UAT and Alpha Testing** help your team understand your prototype's strengths and weaknesses? What did you learn about your users or system from this process?

Conducting UAT and Alpha Testing helped our team identify system weaknesses, usability issues, and areas that needed improvement before presenting the system to external users.

**Think About:** How did creating a **Revision Plan** help your team turn feedback into clear, actionable steps for improvement? How will this prepare you for **Beta Testing or external evaluations** in the next phase?

Creating a revision plan helped our team organize feedback into clear tasks and solutions for improving the system. This prepares us for Beta Testing and external evaluations by making the system more reliable, user-friendly, and ready for actual users and evaluators.